

ORIGINAL ARTICLE

Periostin: A promising biomarker and key mediator of liver fibrosis in patients with chronic hepatitis C

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ABSTRACT

Background and aim: Chronic injury to hepatocytes caused by the hepatitis C virus can lead to fibrosis, necessitating continuous monitoring and follow-up assessments. Elevated periostin levels have been observed in various fibrotic diseases, making it a potential non-invasive biomarker. This study aims to determine the role of periostin serum levels as a fibrosis biomarker in chronic hepatitis C patients, in relation to viral load, antivirals and other baseline liver tests.

Methods: A case-control study at the Liver and Gastrointestinal Hospital in Baghdad involved 60 patients with chronic hepatitis C and 25 healthy controls. We measured serum periostin levels using an enzyme-linked immunosorbent assay (ELISA). We analysed the results in relation to viral loads, liver enzyme levels, ultrasound findings, and fibro-scan results.

Results: The median interquartile range (IQR) of serum periostin was 35.62 ng/L in treated hepatitis C patients and 27.85 ng/L in non-treated patients. In comparison, the control group had a significantly lower median concentration of 13.90 ng/L ($P = 0.001$). Using a periostin cutoff of 37.30 ng/L, 22 patients (36.7%) had an AST/ALT ratio of ≥ 1 , 20 patients (33.3%), had an APRI score of ≥ 0.5 , and 10 patients (16.7%) showed ultrasound evidence of fibrosis with varying fibroscan results (F1-F4). A strong positive correlation ($r = 0.8$) was observed between periostin levels and HCV viral load in non-treated patients.

Conclusions: This study suggests that elevated levels of periostin are associated with active HCV infection and show a potential relationship with liver fibrosis. (www.actabiomedica.it)

Key words: biomarker, direct acting antivirals (DAAs), chronic hepatitis C, liver fibrosis, periostin



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Introduction

The Hepatitis C virus (HCV) is an RNA virus member of the Flaviviridae family (1). It causes inflammation of hepatocytes with high rates of chronicity and leads to a significant health issue such as liver fibrosis, cirrhosis and hepatocellular carcinoma (HCC). An increased mortality related to late-stage HCV liver infection (2) was recorded by The World Health Organization (WHO), which estimated that in 2022, approximately 242,000 deaths were attributed to post hepatitis C cirrhosis and HCC. Moreover, around 50 million people are living with chronic infection, with 1 million new cases emerging each year (1).

In Iraq, the estimated prevalence of HCV viraemia is approximately 78,620 persons (3). While direct-acting antivirals (DAAs) can cure over 95% of those with hepatitis C infection, access to diagnosis and treatment remains a significant challenge (1). Morbidity and mortality associated with liver disease are closely linked to the extent of fibrosis. Thus, it is crucial to evaluate the degree of fibrosis in individuals infected with HCV to inform the timing of treatment, guide screening for complications, and facilitate post-cure monitoring. Although the mechanisms underlying fibrosis in HCV infection are still being explored, it is clear that liver fibrosis often arises as a reparative response to liver injury or inflammation (4). Non-invasive approaches based on the detection and measurement of numerous serum biomarkers were used to examine the degree of long-lasting hepatocyte damage (5,6) and track the progression towards fibrosis. These biomarkers are classified into two categories: direct markers reflecting fibrotic activity and indirect markers indicating deterioration of liver function. Both types may provide invaluable insights into the extent of liver damage (7). The use of noninvasive methods including serum liver biomarkers, offers numerous advantages for HCV patients, including speed, cost-effectiveness, and the ability to conduct repeated tests (7,8).

Periostin is recognized as a non-structural extracellular secretory protein implicated in fibrogenic processes. In a healthy adult liver, its expression is low. It becomes overexpressed in response to tissue injury or fibrous stimuli across multiple organs including lung, kidney, heart, and liver (9–11). It is secreted by triggered

hepatic stellate cells (HSCs) in response to transforming growth factor beta-1 (TGF- β 1). Periostin not only enhances TGF- β activity but also appears to engage in a reciprocal regulatory relationship with it, indicating its complicated contribution to the progression of liver fibrosis. Of note, in HCV where inflammation may taper off after viral cure, periostin may uniquely reflect residual extracellular matrix structure deterioration. Elevated serum levels of periostin have been detected in patients with both acute and chronic hepatitis in comparison to healthy individuals, supporting its association with fibrogenic processes rather than inflammation alone (12). However, studies have shown that serum periostin levels do not change significantly during DAA therapy (13). Despite the widespread use of direct-acting antivirals and high sustained virological response rates, the behavior of fibrosis-related biomarkers such as periostin following viral clearance is still not fully understood, raising questions about its role in fibrosis progression during treatment. Overall, while mechanistic data support a role for periostin in liver fibrosis in general, direct evidence implicating periostin specifically in HCV-mediated fibrogenesis is still limited. This observation highlights the need for further research to clarify the precise function of periostin in HCV-related liver fibrosis. This preliminary study explores the association of periostin with fibrosis-related parameters and its potential role as a non-invasive biomarker in patients with chronic hepatitis C, providing a deeper understanding of its roles in disease progression.

Material and methods

A case-control study was conducted in the Department of Microbiology, College of Medicine, University of Baghdad. The ethical approval for this study was obtained from the Ethics Committee of the Faculty of College of Medicine at University of Baghdad, in accordance with IRB: reference No. 0245 in 24th of September 2024. Informed verbal and written consent was obtained from all participants to participate in this study. The sampling and methods extended from September 2024 to March 2025. The study included 60 patients with hepatitis C infection compared to 25

healthy control participants collected randomly from friends and relatives who don't suffer from any disease. Blood sample were collected from patient who were attending Gastroenterology and Hepatology Center in Baghdad Teaching Hospital. Participants were included in the study if they had been diagnosed with hepatitis C for six months or longer and had completed direct-acting antiviral therapy, being three months into their recovery. Participants were not enrolled if they had co-infections with other viral hepatitis or HIV, liver fibrosis secondary to other medical conditions, or a history of allergic, autoimmune, or respiratory disorders. Individuals with recent COVID-19 infection and post-COVID complications, those undergoing hemodialysis, pregnant women, and participants with alcohol use disorder were also not recruited. The samples were collected after the agreement of each patient to be part of this study. Then classified into two groups as follows: Twenty-six newly diagnosed hepatitis C patient before receiving any treatment and thirty-four patients who were treated with one of the Direct-Acting Antivirals (DAAs) with daily recommended dose for three months, including Ecluzac (velpatasvir 100mg/sofosbuvir 400mg) or Harvoni (90/400mg of Ledipasvir /sofosbuvir). Additionally, one case received VOSEVI (Sofosbuvir 400mg /velpatasvir 100 mg/voxilaprevir 100 mg). All cases were diagnosed through proper investigations and were under the care of gastroenterology consultants, using the ELISA method along with confirmatory HCV-RNA real-time polymerase chain reaction to confirm the diagnosis and measure the viral load. After the agreement of each patient to participate in the study, they were subjected to a data form which included socio-demographic data such as age and sex. Complete medical history was taken from each patient. All registered investigations were recorded (clinical and laboratory data). Five ml of blood was drawn aseptically, then placed in gel tubes, separated as serum by centrifugation for 5 minutes at 3000 rpm, and divided into Eppendorf tubes of 250 μ L and stored at -20°C till examination. Each Eppendorf tube of serum was used once to avoid sample deterioration due to repeated freezing and thawing cycles. All serum and reagents were allowed to stand at room temperature before use in the test. Periostin concentration was calculated by ELISA Biotech system (ELISA reader

spectrophotometer, centrifuge and automated washer) after serum processing steps written in the leaflets of Quick Step Human Periostin (POSTN) ELISA kit (SunLong Biotech Co., LTD, China, Cat. NO. QS2267Hu).

Statistical analysis

The results were organized into tables and figures after being calculated by the use of Microsoft Excel sheet 2016 and loaded into IBM SPSSV26 statistical program.

Periostin concentration was expressed as mean $\pm$ standard deviation (SD) or as median with interquartile range (IQR).

Comparisons among more than two groups were conducted using one-way analysis of variance (ANOVA). Associations between categorical variables were analyzed utilizing the Chi-square test or Fisher's exact test when expected cell counts were less than five. A p-value of less than 0.05 was deemed statistically significant. The diagnostic performance of the studied biomarker was rigorously evaluated by calculating sensitivity, specificity, and Youden's index. Furthermore, receiver operating characteristic (ROC) curves were constructed, and the corresponding areas under the curve (AUC) values were determined. Also, a regression curve was plotted, and the correlation coefficient (r value) was calculated.

Results

Demographic distribution of study groups

A total of 85 participants, who were categorized into three main groups: 34 (40%) individuals were HCV-treated patients, 26 (30.6%) were non-treated HCV patients, and 25 (29.4%) were apparently healthy participants, serving as the control group.

Table-1 illustrates that the mean age $\pm$ SD of the non-treated participants was 36.3 $\pm$ 19.5 years. In comparison, the treated patients had a mean age $\pm$ SD of 42.0 $\pm$ 17.5 years, while the control group had a mean age $\pm$ SD of 38.5 $\pm$ 13.1 years. The p-value for the comparison between patients and controls was 0.672, whereas the p-value for the comparison between treated and non-treated patients was 0.171. Data were detailed in Table 1.

Serum levels of periostin concentration among study groups

As presented in Table 2, the median interquartile range (IQR) concentration of periostin among treated patients was observed to be 35.62 ng/l, while the median concentration for non-treated patients was recorded at 27.85 ng/l. It is noteworthy that no statistically significant difference was identified between the treated and non-treated groups ($p = 0.865$). In comparison, the control group exhibited a median IQR concentration of periostin of 13.9 ng/l, indicating a highly significant difference ($p=0.0001$) compared to both patient groups.

ROC Curve of periostin serum concentration among patients and controls

The marker would provide an almost good test as illustrated in Figure 1. The area under the curve (AUC) was 0.969 with strong significance ($p = 0.000$) among study groups (cases and control).

The optimum cut-off level of periostin concentration was 37.30 ng/l with a sensitivity and specificity of 92 % and 88 % respectively measured by Youden index.

Number of Periostin positive versus negative cases in the study

Based on the cut-off value of 37.3 ng/l, the serum concentration of periostin among all participants is divided into two categories: positive ($>37.3\text{ng/l}$) and negative ($\leq 37.3\text{ng/l}$). As it appears in Table 3, the serum periostin concentrations were observed in 31 (36.5%) of treated cases and 24 (28.2%) in non-treated cases with ($p=0.87$). Conversely, the highest percentage of negative periostin concentrations was found in the control group 22 (25.9%), which demonstrates a highly statistically significant difference compared to all patients, with a $p\text{-value} = 0.0001$.

Table 1. Demographic distribution of study groups.

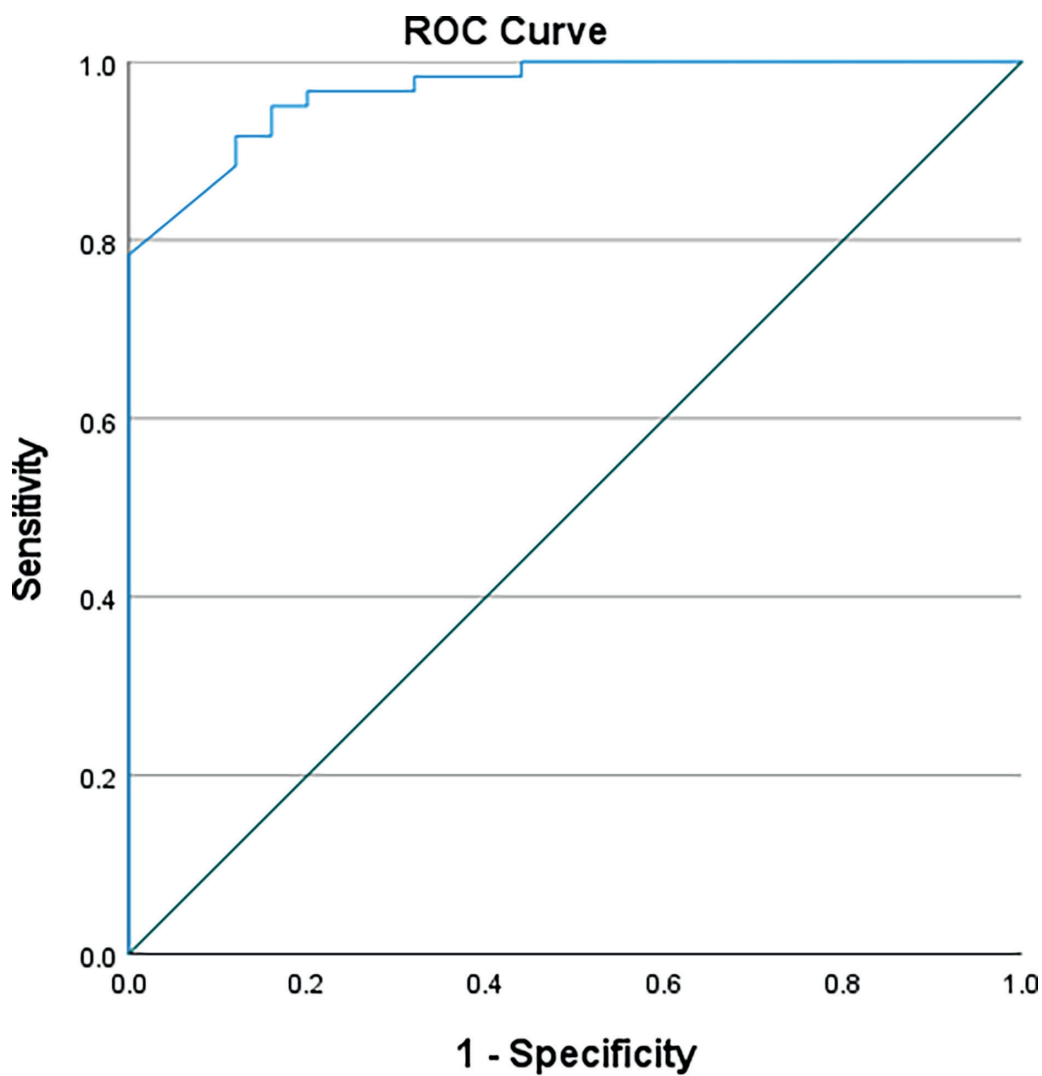
Characters	HCV – patients		Control group	P- value
	Treated	Non-treated		
Age (year)				
Range	(12-80)	(10-80)	(20-62)	
Mean $\pm$ SD	(42.0 $\pm$ 17.5)	(36.3 $\pm$ 19.5)	(38.5 $\pm$ 13.1)	
Sex: NO (%)				*0.672
Female	19(55.9%)	16(61.5%)	12(40%)	**0.171
Male	15 (44.1%)	10 (38.5%)	18 (60%)	
F:M ratio	1.3:1	1.6:1	0.7:1	
Total No. (%)	34 (40%)	26 (30.6%)	25(29.4 %)	

Abbreviations: SD: standard deviation; NO: number; F:M ratio: female to male ratio. *P-value of patient/control; ** p-value of Treated / Non-treated patient

Table 2. Serum level of periostin concentration among study groups.

Groups		Periostin concentration (ng/l)		P-value
		Mean $\pm$ SD	Median IQR(Q3-Q1)	
Patients	Treated	64.75 $\pm$ 25.00	35.62(29.27 – 43.65)	0.865
	Non-treated	66.17 $\pm$ 39.37	27.85(19.72– 41.87)	
Control		27.56 $\pm$ 9.04	13.90(34.80 – 20.90)	0.0001

Abbreviations: IQR: interquartile range.



Diagonal segments are produced by ties.

Figure 1. ROC Curve of periostin serum concentration among patients and controls.

Table 3. Number of positive versus negative periostin results in the study.

Groups (Total NO.)		Periostin concentration (ng/l) NO. (%)		p-value
		Positive	Negative	
Patients (60)	Treated (34)	31(36.5)	3(3.5)	*0.87
	Non-treated (26)	24(28.2)	2(2.4)	
Control (25)		3(3.5)	22(25.9)	**0.0001

* p-value by Chi square test between patient/control group; ** p-value by Fisher's exact test to treated / non-treated patients

Association between types of DAAs and periostin serum concentration in treated HCV cases

In Table 4, it was noted that the majority of the patients (82%) were treated with Cluza-c (velpatasvir 100mg/sofosbuvir 400mg) and tested positive for high periostin concentration. However, no statistically significant difference was observed when compared to the negative periostin results. Additionally, four cases were treated with HARVONI (ledipasvir 90mg/sofosbuvir 400mg), and three of these cases were positive according to periostin cut-off value. Similar to Cluza-c, there was no statistically significant difference compared to the negative group.

Only one case received VOSEVI treatment (sofosbuvir 400mg /velpatasvir 100 mg/voxilaprevir 100 mg) was given a negative result of periostin level. Again, no statistically significant difference was observed compared to the negative group.

Impact of periostin concentration results on baseline parameters in HC patients

The relationship between periostin concentration and baseline parameters in patients with positive and negative results was analyzed in this study. As summarized in table-5, twenty-one patients with positive periostin and abnormal serum bilirubin levels were formed 35% of total cases.

In addition, 33 (55%) had an AST/ALT ratio of <1 compared to 22 (36.7%) of total patients who were found to have a high ratio of (≥ 1) with a positive

level of periostin. Additionally, 20 (33.3%) of patients showed an APRI score ≥ 0.5 with positive periostin concentration.

According to liver U/S and Fibro-scan results, most of the patients with a positive periostin level had significantly more than one sonographic finding, and various degrees of fibrosis (F1-F4) compared to patients with a negative periostin level.

The correlation between serum level of periostin concentration with viral load in study HCV-non-treated cases

The Simple Scatter/ Dot graph in Figure 2 demonstrates a positive linear correlation coefficient ($r = 0.8$; $P=0.000$) between periostin concentration and HCV-viral load in non-treated patients rather than treated cases as the viral load of the latter was undetectable at less than 10 IU/mL, while, in non-treated cases it ranged from 290 to 9.7×10^3 IU/mL with a mean $\pm$ SD of $2.4 \times 10^3 \pm 2.8 \times 10^3$ IU/mL. This indicates that as HCV viral load increases in a patient's blood, the serum concentration level of periostin also tends to rise. This relationship is considered statistically strong.

Discussion

Serum levels of periostin concentration among study groups

This study found that serum periostin concentration was higher in patients with chronic hepatitis C

Table 4. Number of positive versus negative periostin serum levels in association with DAAs type.

Drug used		Periostin concentration(ng/l) No (%)		P- value
		Positive	Negative	
velpatasvir 100mg/sofosbuvir 400mg	Yes	28(82)	2(6)	0.32
	No	3(9)	1(3)	
ledipasvir 90mg/sofosbuvir 400mg	Yes	3(9)	1(3)	0.32
	No	28(82)	2(6)	
sofosbuvir 400mg /velpatasvir 100 mg/voxilaprevir 100 mg	Yes	0	1(3)	0.08
	No	31(91)	2(6)	
Total No. (34)		31(%)	3(%)	

compared with healthy controls, and that cases who had completed DAA therapy and were three months into recovery showed a partial decline in periostin compared with untreated CHC patients, but did not always reach control levels. These preliminary observations are consistent with the proposed role of periostin as a matricellular mediator of liver fibrogenesis. Periostin is produced by activated hepatic stellate cells and promotes extracellular matrix deposition and tissue hardening through interaction with lysyl oxidase family members that mediate collagen cross-linking, stabilising fibrillar collagen and stiffening the matrix, along with TGF- β signalling, processes that underpin fibrosis development (14).

Comparison with previous studies provides a mixed but informative prospective. Several experimental

and clinical reports have reported elevated periostin expression in fibrotic or inflamed liver tissue and in the serum of patients with acute or chronic hepatitis, these findings suggest that periostin may act as mediator (not just a marker) of fibrosis rather than a passive bystander. For example, Huang et al. described higher serum periostin in patients with hepatitis compared with healthy controls and demonstrated an association between periostin and TGF- β expression (13).

Several studies have investigated the dynamics of periostin in the context of CHC and its response to antiviral therapy such a study by in 2023 found that while DAA therapy significantly decreased serum levels of TGF- β , a key pro-fibrotic mediator, there was no significant change in periostin levels during the treatment period. This suggests that periostin regulation

Table 5. The relationship between serum levels of periostin and baseline parameters in studied cases.

Patient's Parameters (normal value)	Periostin concentration(ng/l) NO (%)		P-value
	Positive	Negative	
Serum bilirubin (0.2- 1.0 mg/dl) * (>1.0 mg/dl) **	34 (56.7)	5 (8.3)	0.15
	21 (35)	0	
AST/ ALT ratio (<1) * ($\geq$ 1) **	33 (55)	5(8.3)	0.14
	22 (36.7)	0	
APRI (<0.5) * ($\geq$ 0.5) **	35 (58.3)	4(6.7)	0.6
	20 (33.3)	1(1.7)	
***U/S No. (%)			0.002
Normal liver span (13.45 cm)	22(36.7)	2(3.3)	
Hepatomegaly	16(26.7)	0	
Fatty liver	8(13.3)	0	
fibrosis	10(16.7)	0	
Cirrhosis	2(3.3)	0	
N/A	17(28)	3(5)	
Fibroscan results: No. (%)			0.058
Mild fibrosis (F1)	11(18.3)	1(1.7)	
Moderate fibrosis (F2)	5(8.3)	0	
Severe fibrosis (F3)	8(13.3)	0	
Cirrhosis (F4)	3(5)	0	
N/A	28(46.7)	4(6.7)	

*Normal, ** Abnormal; *** Most patients had more than one ultrasound (U/S) finding. AST: aspartate aminotransferase; ALT: alanine aminotransferase; APRI: AST to Platelet Ratio Index (APRI); U/S: ultrasound; N/A: Non available.

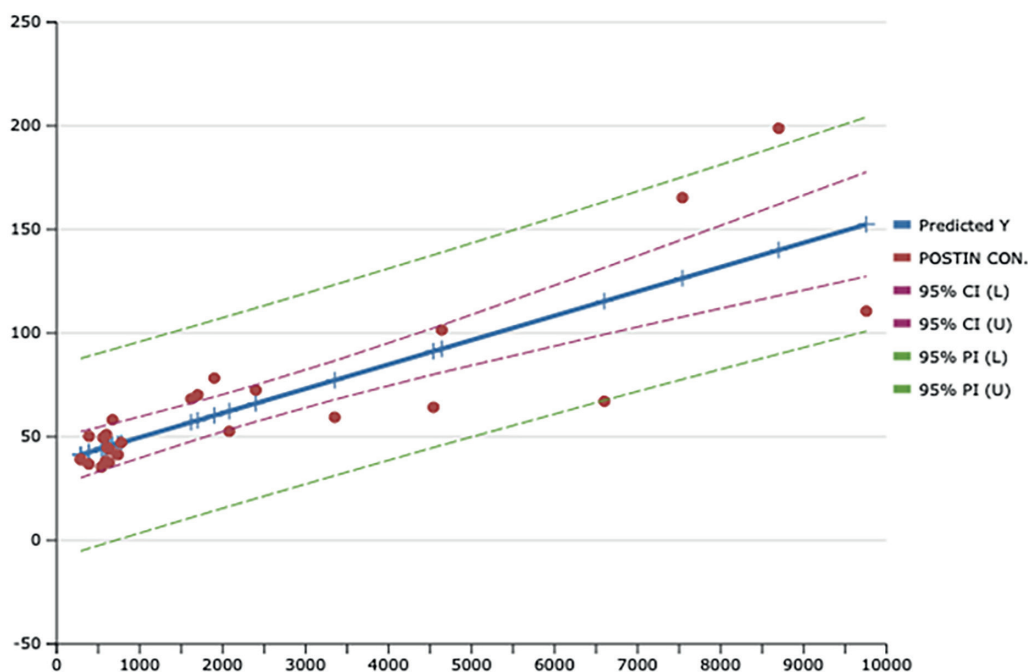


Figure 2. The correlation between serum level of periostin concentration with viral load in study HCV non-treated cases.

may be influenced by factors beyond TGF- β , such as IL-4 and IL-13, or that the duration of follow-up post-treatment was insufficient to observe changes in periostin levels (12). A study in 2019 highlighted the potential role of periostin in liver fibrosis, noting its involvement in the progression of chronic liver diseases. Their findings underscore the relevance of periostin as a potential biomarker for monitoring fibrotic changes in the liver (15). Viral clearance via DAA reduces pro-fibrotic signaling (for example, lowers TGF- β and inflammatory mediators) relatively quickly, but reversal of matrix crosslinking and of matricellular proteins such as periostin may require a longer recovery interval or adjunctive anti-fibrotic interventions (16).

ROC Curve of periostin serum concentration among patients and control

In the current study, an exploratory serum periostin cutoff value of 37.30 ng/l was identified for patients with chronic hepatitis C. Periostin levels above this threshold presented a sensitivity of 92% and specificity of 88% within our cohort, suggesting a potential

association with more advanced fibrosis-related stages. Although periostin demonstrated a high AUC in this cohort, given the limited sample size and the absence of external validation, the proposed periostin cut-off values should therefore be considered exploratory and these findings should be interpreted cautiously but are broadly consistent with previous reports examining periostin involvement in liver fibrogenesis (16).

The majority of chronic hepatitis C patients were periostin-positive, whereas most healthy controls were periostin-negative and therefore showed a highly statistically significant difference. This finding reflects the role of periostin as a potential biomarker of liver fibrosis and ongoing hepatocyte structure remodeling (15). Elevated periostin in patients likely reflects activation of hepatic stellate cells and enhanced extracellular matrix deposition, processes that are not present in healthy individuals (17). Similar observations have been reported in other studies of liver disease, where serum periostin levels were significantly higher in patients with chronic liver conditions compared to healthy controls, supporting its association with fibrogenic activity (18). These results underscore that

periostin may help to differentiate patients with ongoing fibrogenesis from healthy subjects.

Association between types of DAAs and periostin serum concentration in treated HCV cases

In terms of DAA types, there was no significant association between the type of DAA therapy and periostin positivity. Although many patients received Cluza-C (velpatasvir 100 mg/sofosbuvir 400 mg), the proportion of periostin-positive treated-patients was similar across different treatment regimens. This suggests that periostin expression may be independent of the specific antiviral used, reflecting ongoing fibrogenic or remodeling activity rather than the effect of the drug itself or viral clearance. These findings are consistent with previous studies showing that periostin levels may remain elevated after DAA therapy regardless of the drug types, and many studies support periostin's role as a treatment-independent marker of hepatic fibrosis(19, 20). Sound judgment on the effect of DAA treatments of periostin burden should be evaluated further on large-scale groups, along with longer periods.

Impact of periostin concentration results on baseline parameters in patients on treatment

The relationship between periostin concentration and baseline parameters in patients with positive and negative results was analysed in this study, however, our results didn't indicate an association between periostin and bilirubin in patients. This suggests that periostin, operates independently of bilirubin, which reflects hepatocellular excretory dysfunction (15). Similar results were reported in non alcoholic fibrotic liver disease, where periostin have no significant association with bilirubin (17,18). These findings indicate that periostin captures fibrotic activity earlier than bilirubin changes. Clinically, periostin may complement bilirubin, providing insight into fibrosis progression before functional decline becomes evident.

Serum periostin concentration in this study also showed no significant association with the AST/ALT ratio or APRI index. The highest percentages were

observed in patients with positive periostin levels but within the normal parameter range, mainly because 34 participants were treated patients who had normalized biochemical parameters. Therefore, there was no significant association between periostin concentration and the liver parameters studied. Or this finding indicates that periostin reflects fibrogenesis independently of biochemical injury markers or platelet-based fibrosis scores. Similar observations were noticed in the both study where periostin did not consistently align with non-invasive indices (19). Therefore, periostin may provide complementary information to conventional indices in exploring its role in liver disease (15).

The correlation between serum level of periostin concentration with viral load in study HCV-non-treated cases

In our study of non-treated chronic hepatitis patients, we observed a strong positive correlation between serum periostin concentration and HCV viral load meaning patients with higher HCV RNA had higher periostin levels. This finding suggests that periostin may reflect the degree of viral replication and associated fibrogenic activity (15). Direct experimental evidence linking periostin and viral load in HCV is limited. A study reviewed periostin in chronic liver disease and showed higher periostin with fibrosis severity, but did not derive cutoffs for viral load (15).

Animal and translational models study showed periostin induction under fibrogenic stress and reduction with renin-angiotensin system blockade, supporting the biological plausibility (17). Other studies measured circulating periostin in liver disease cohorts and they confirmed periostin elevation in chronic disease but found heterogeneous correlations with non-invasive fibrosis scores (18,21). Interestingly, our findings in non-treated patients contrast with evidence from studies of DAA-treated HCV cases. For example, in the study fibrogenic signals persist in DAA-treated HCV patients after sustained virological response, markers such as TGF- β 1 remained elevated even after HCV RNA became undetectable (20). This suggests that although viral replication is an important driver of periostin expression, fibrogenic pathways may persist independently of viral load once

they are established. Periostin may reflect the austerity of fibrous activity. Thus, it may be part of the pathophysiology that links chronic hepatitis C virus to viral persistence, fibrosis, and further HCC risk. Additional evidence supporting periostin as a marker associated with fibrogenic activity comes from other study that reported significantly elevated serum periostin levels in patients with non-alcoholic fatty liver disease compared with healthy controls. These findings suggest that periostin may be linked to liver injury across diverse etiologies (17). Unlike our findings where periostin level closely tracked HCV viral load, it showed periostin elevation in the absence of viral replication, indicating that periostin reflects a broader fibrogenic process, not just viral activity. Accordingly, periostin may have a dual relevance: as a dynamic marker associated with viral activity in non-treated HCV infection and as a potential biological marker of ongoing or residual fibrotic process in post-SVR or non-viral liver diseases settings. The association between elevated periostin level and higher prevalence of HCC within HCV cohort may suggest that periostin reflects an advanced micro-fibrous environment. Accordingly, in HCV patients, periostin may help identify HCV patients with more pronounced ECM remodelling even after virus has been cleared. Several limitations must be recognized. The relatively small sample size, especially in subgroup analyses, limits statistical power and may amplify effect estimates. The cross-sectional design precludes assessment of temporal or causal relationships. In addition, liver tissue histology was not available, and the assessment of fibrosis relied on heterogeneous non-invasive methods, which may cause a variation in measurement variability. Accordingly, our findings should be interpreted as exploratory and require confirmation in larger, prospective studies with standardized fibrosis assessment.

Conclusions

This study provides preliminary evidence to support the association between serum periostin levels, viral activity, and fibrosis-related parameters in chronic hepatitis C. While periostin shows potential as a non-invasive biomarker, particularly in non-treated

patients, these results should be interpreted with caution. Larger, future studies involving standardized fibrosis evaluation and longitudinal follow-up are needed to elucidate the biological significance of periostin in the post-DAA era.

Recommendations

Implementing long-term follow-up to observe changes in periostin levels in serum and liver tissues over time and their correlation with liver function tests and histological assessments. Conducting studies with larger and more diverse patient populations to validate periostin as robust non-invasive biomarker of liver fibrosis in CHC. Regarding the therapeutic targeting of periostin, although promising in preclinical models (mostly of non-viral fibrogenesis), translation into HCV-specific patient therapies is still speculative.

Ethic approval: Ethics Committee of the Faculty of College of Medicine at University of Baghdad, IRB: reference No. 0245 in 24th of September 2024.

Conflict of interest: Each author declares of non-interest regarding any commercial associations as consultancies, stock ownership, equity interest, patent/ licensing arrangement to the company that produced the assay kit (SunLong Biotech Co.,LTD, China). That might lead to a conflict of interest in the submitted article.

Authors contribution: Abrar Albiram collected patient samples and conducted test procedures, while Shatha Abdullah was responsible for the study design. Both authors collaborated on data analysis, writing and manuscript revision.

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